GATE

CRASH COURSE

ALL BRANCHES

Engineering Mathematics

Lecture No. 05

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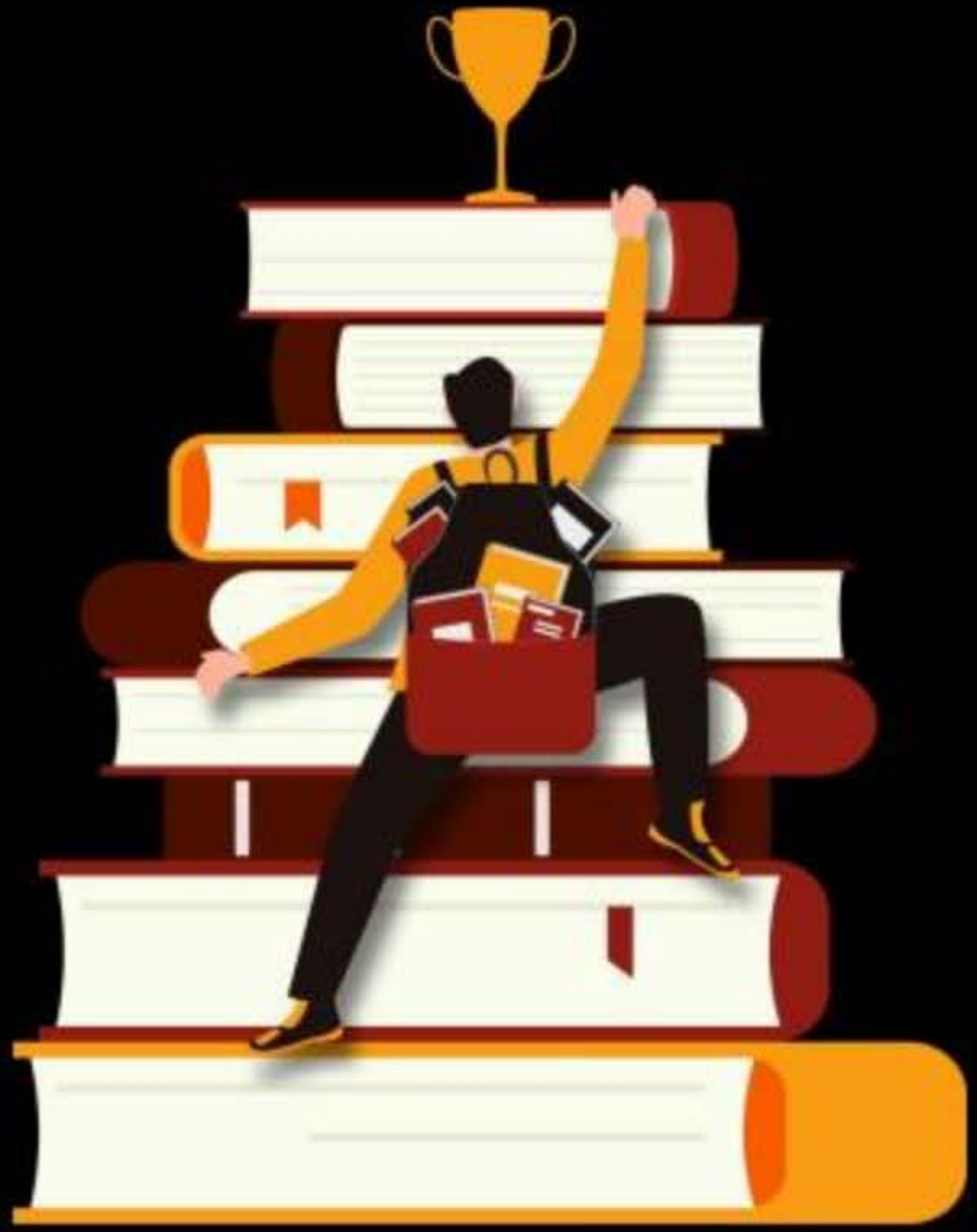
Topics *to be covered*

PROB & STATS (Part-2)



COUNTING PRINCIPLE

Fundamental Principle of Addition → If we have to perform only one of the job at a time out of n jobs then use this principle.

Keywords : "Either or / only one / Any one"

Fundamental Principle of Multiplication → If we have to perform all the jobs at a time out of n jobs then use this principle.

Keywords: "All / Both / AND"

Q: How many 3 letter words can be formed (with/w/o meaning) using Eng. alphabets.

① if Repetition is Not allowed = ? = $\underset{P_1}{\underline{26 \text{ ways}}} \times \underset{P_2}{\underline{25 \text{ ways}}} \times \underset{P_3}{\underline{24 \text{ ways}}} = ? = {}^{26}P_3$

② if " is allowed = ? = $\underline{26 \text{ ways}} \times \underline{26 \text{ ways}} \times \underline{26 \text{ ways}} = ?$

Q How many 4 digit Nos can be formed using 1, 3, 5, 7, 9

① RNA = ? $= \underset{P_1}{\underline{5 \text{ ways}}} \times \underset{P_2}{\underline{4 \text{ ways}}} \times \underset{P_3}{\underline{3 \text{ ways}}} \times \underset{P_4}{\underline{2 \text{ ways}}} = {}^{5}P_4$

② RA = ? $= \underset{P_1}{\underline{5 \text{ ways}}} \times \underset{P_2}{\underline{5 \text{ ways}}} \times \underset{P_3}{\underline{5 \text{ ways}}} \times \underset{P_4}{\underline{5 \text{ ways}}} =$

In a test there are eight questions, in which four have three possible answers, three has two possible answers each and one question has five possible answers. The total number of possible answers will be?

(a) 2880 (b) 78 (c) 94 (d) 3240

Total possible Ans $= \underset{Q_1}{3} \times \underset{Q_2}{3} \times \underset{Q_3}{3} \times \underset{Q_4}{3} \times \underset{Q_5}{2} \times \underset{Q_6}{2} \times \underset{Q_7}{2} \times \underset{Q_8}{5}$ ways

$= 3^4 \times 2^3 \times 5 = 3240$ (d)

(*) If counting is Based on Selection only use ${}^{n}C_{r}$

(*) " " " " " Selection as well as Arrangement also use ${}^{n}P_{r}$.

$$\because \boxed{{}^{n}P_{r} = {}^{n}C_{r} \times r!}$$

Perm — Selection — Arrangement

But these Results are applicable only when RNA

(*) If RA then only use Multi Rule

The greatest possible number of points of intersection of 8 straight lines and 4 circles is

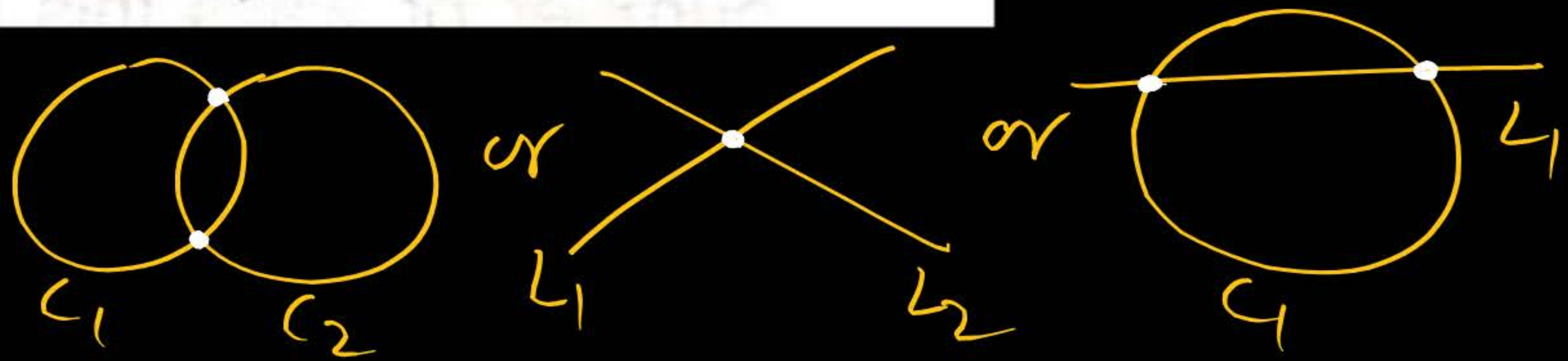
(a) 32

(b) 64

(c) 76

(d) 104

Possible cases are



Req Points of intersection are $= {}^4C_2 \times 2 + {}^8C_2 \times 1 + {}^4C_1 \times {}^8C_1 \times 2$

$= 104$ (d)

H.Q ⇒ SALADS are made from one or more eatables. then how many different salads can be made using onion, Tomato, Reddish, Carrot, Beetroot.

we can make salad either by selecting

= 1V or 2V or 3V or 4V or 5V

$$= {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5$$

$$= 5 + 10 + 10 + 5 + 1 = 31 \text{ Salads}$$

In a test there are three multiple choice questions having four choices each. Number of sequences in which a student can fail to get all answers correct is

(a) 11 (b) 15 (c) 80 (d) 63

Total possible sequences of answers

$$= \frac{4 \text{ ways}}{Q_1} \times \frac{4 \text{ ways}}{Q_2} \times \frac{4 \text{ ways}}{Q_3} = 64 \text{ ways}.$$

in which only one sequence has all correct answers.

Req Ans $= 64 - 1 = 63$ ways. (d) ✓

In a chess competition involving some boys and girls of a school, every student had to play exactly one game with every other student. It was found that in 45 games both the players were girls and in 190 games both the players were boy. The number of games in which one player was a boy and other was girl is

(a) 200 (b) 216 (c) 235 (d) 256

ATQ, ${}^{B}C_2 = 190$ & ${}^{G}C_2 = 45$

$\boxed{{}^{n}C_2 = \frac{n(n-1)}{2}}$ Learn.

$\frac{B(B-1)}{2} = 190$

$B(B-1) = 380$

$B(B-1) = 20 \times 19$

$B(B-1) = 20(20-1)$

$B = 20$

$\frac{G(G-1)}{2} = 45$

$G(G-1) = 90$

$= 10 \times 9$

$G(G-1) = 10(10-1)$

$G = 10$

eg ${}^{6}C_2 = \frac{6 \times 5}{2} = 15.$

So Ans $= {}^{B}C_1 \times {}^{G}C_1$

$= {}^{20}C_1 \times {}^{10}C_1 = 200.$

In a population of N families, 50% of the families have three children, 30% of the families have two children and the remaining families have one child. What is the probability that a randomly picked child belongs to a family with two children?

(a) 3/23
(b) 6/23
(c) 3/10
(d) 3/5

Sol (b)

Let $N = 100$ families

Total No. of Children $= 50\times3 + 30\times2 + 20\times1 = 230$.

Fav. No. of Children $= 30\times2 = 60$

So Req Prob $= \frac{\text{Fav}}{\text{Total}} = \frac{{}^{60}C_1}{{}^{230}C_1} = \frac{60}{230} = \frac{6}{23}$

M-II Total Kids $= \left(\frac{50N}{100}\times 3 + \frac{30N}{100}\times 2 + \frac{20N}{100}\times 1\right)$

Fav Kids $= \frac{30N}{100}\times 2$

$$\text{Prob} = \frac{\text{Fav}}{\text{Total}} = \frac{\frac{30N}{100}\times 2}{\frac{50N}{100}\times 3 + \frac{30N}{100}\times 2 + \frac{20N}{100}\times 1} = \frac{6}{23}$$

The number of times the digit '3' will be written when listing the integers from 1 to 1000, is

(a) 269
(b) 271
(c) 302
(d) 300

$\underbrace{0, 1, 2, 3 \ldots, 8, 9}_{10 \text{ digits}}$

Various possible Cases = '3' is occurring once $= \left({}^3C_1 \times \underline{1} \times \underline{9} \times \underline{9}\right) \times 1$

or

'3' is " twice $= \left({}^3C_2 \times \underline{1} \times \underline{1} \times \underline{9}\right) \times 2$

or

'3' is " thrice $= \left({}^3C_3 \times \underline{1} \times \underline{1} \times \underline{1}\right) \times 3$

Req Ans $= 3 \times 81 + 3 \times 18 + 1 \times 3 = 300$.

(d) ✓

Case I: 3 $\underline{9} \times \underline{9}$ or $\underline{9}$ 3 $\underline{9}$ or $\underline{9} \times \underline{9}$ 3 $\Rightarrow$ 3 case $\times 81 \times 1$

Case II: 3 3 $\underline{9}$ or 3 $\underline{9}$ 3 or $\underline{9}$ 3 3 $\Rightarrow$ 3 case $\times 9 \times 2$

Case III: 3 3 3 $\Longrightarrow$ 1 Case $\times 3$ times

Arrangement:

How many three letter computer passwords can be formed with at least one symmetric letter such that Repetition not allowed. It is given that symmetric letters are A, H, I, M, O, T, U, V, W, X, ~~Z~~. Y = 11 symm letter

(a) 990 (b) 2730 (c) 12870 (d) 1560000

No. of passwords in which at least one symm letter is there

= Total 3 letter passwords − Number of passwords with NO symm letter

M-I $= \left(\underset{P_1}{26} \times \underset{P_2}{25} \times \underset{P_3}{24}\right) - \left(\underset{P_1}{15} \times \underset{P_2}{14} \times \underset{P_3}{13}\right)$

M-II $= {}^{26}P_3 - {}^{15}P_3 =$ (c)

Various Cases:-

$$(\text{No symm letter}) \text{ or } \underbrace{(\text{1 symm letter}) \text{ or } (\text{2 symm letter}) \text{ or } (\text{3 symm letter})}_{\text{At symm one symm letter}} = \text{Total}$$

$$(X=0) + \underbrace{(X=1) + (X=2) + (X=3)}_{X \geq 1} = \text{Total}$$

$$\boxed{P(X \geq 1) = 1 - P(X=0)}$$

Dearrangements → when No one goes at Right place assigned for him then such types of arrangements are called Dearrangements.

If there are n persons & n directed places. then

Total No. of arrangements $= n!$

RNA

All Correct " $= 1$

All wrong " $= n!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\cdots+(-1)^n\frac{1}{n!}\right]$

for $n=3$, $D_3 = 3!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}\right) = 2$

for $n=4$, $D_4 = \cdots\cdots = 9$

for $n=5$, $D_5 = \cdots\cdots = 44$

HWQ There are 5 Balls & 5 distinct cells. Then find the number

Gate of arrangements in which Ball B_i is not placed in cell C_i $\forall i$ & No cell Remains empty.

Dearrangements $= D_5 = 44$

(a) 44

(b) 1

(c) 119

(d) 120

(a) ✓

Suppose that each of three men at a party throws his hat into the centre of the room. The hats are first mixed up and then each man randomly selects a hat. What is the probability that none of the three men selects his own hat?

(a) $\frac{1}{6}$

(b) $\frac{1}{3}$

(c) $\frac{2}{3}$

(d) 0

Total arrangements of 3 Hats for 3 persons $= 3! = 6$ (RNA).

$$= \begin{pmatrix} P_1 & P_2 & P_3 \\ H_1 & H_2 & H_3 \end{pmatrix}, \begin{pmatrix} P_1 & P_2 & P_3 \\ H_1 & H_3 & H_2 \end{pmatrix}, \begin{pmatrix} P_1 & P_2 & P_3 \\ H_2 & H_1 & H_3 \end{pmatrix}, \begin{pmatrix} P_1 & P_2 & P_3 \\ H_2 & H_3 & H_1 \end{pmatrix}, \begin{pmatrix} P_1 & P_2 & P_3 \\ H_3 & H_1 & H_2 \end{pmatrix}, \begin{pmatrix} P_1 & P_2 & P_3 \\ H_3 & H_2 & H_1 \end{pmatrix}$$

All correct arrangement $= 1$

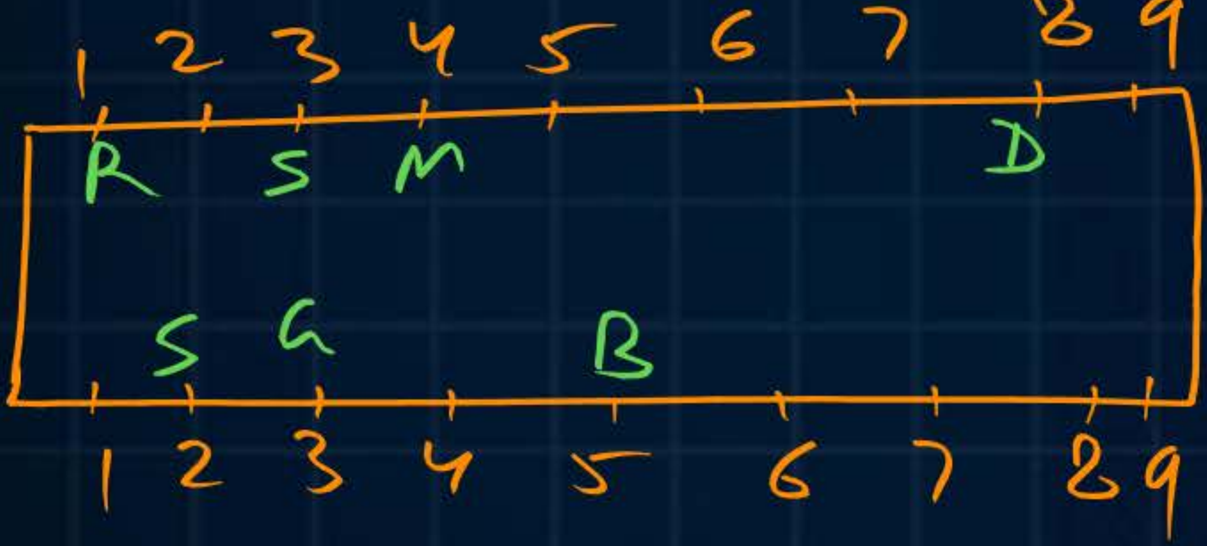
$\boxed{\text{All wrong arrangements}} = 2$

$$P(\text{all wrong Arrangements}) = \frac{2}{6} = \frac{1}{3}$$

(b) ✓

18 guests have to be seated, half on each side of a long table. Four particular guests desired to sit on one particular and three others on the other side, then how many seating arrangements can be made?

(a) ${}^{18}C_4 . {}^{14}C_3 . 9! . 9!$ (b) ${}^{2}C_1 . {}^{9}P_4 . {}^{9}P_3 . 11!$ (c) ${}^{9}P_4 . {}^{9}P_3 . 11!$ (d) ${}^{2}C_1 . \frac{9!}{4!} . \frac{9!}{3!}$



$$\text{Req Arr} = \left({}^{9}C_4 \times 4!\right) \times \left({}^{9}C_3 \times 3!\right) \times {}^{11}C_{11} \times 11!$$

$$= {}^{9}P_4 \times {}^{9}P_3 \times 11!$$

$\left(\because {}^{n}C_r \times r! = {}^{n}P_r\right)$

eg $\left({}^{9}C_3 \times 3! = {}^{9}P_3\right)$

(c) ✓

(*) if side was not particular then

$$\text{Arr} = \left({}^{2}C_1 \times {}^{9}P_4\right) \times \left({}^{1}C_1 \times {}^{9}P_3\right) \times 11!$$

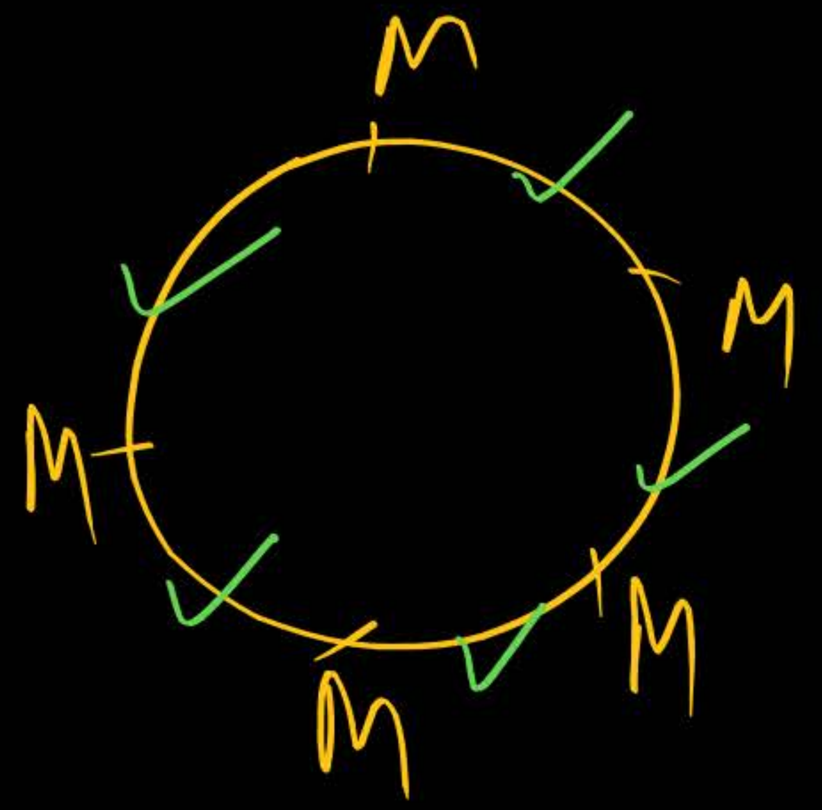
(b) ✓

There are 5 gentlemen and four ladies to dine at a round table. In how many ways can they seat themselves so that no two ladies are together?

Arrangement.

(a) 3280 (b) 2880 (c) 2080 (d) 2480

No two ladies are together = first arrange Males in circle.



$$= (5-1)! \times ({}^{5}C_{4} \times 4!)$$

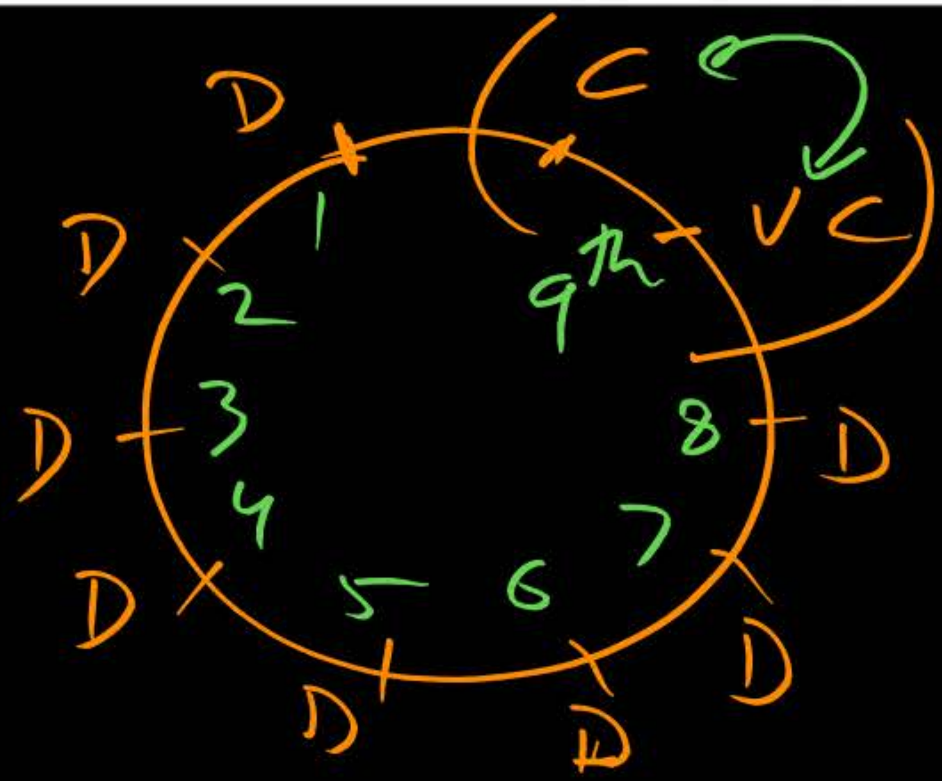
$$= 4! \times {}^{5}P_{4} = 2880$$

(b)

Arrangement

In how many ways can 8 Directors, Vice-Chairman & Chairman of a firm be seated at a round table, if the Chairman has to sit between Vice-Chairman & Director

(a) $2 \times 9!$ (b) $2 \times 8!$ (c) $2 \times 7!$ (d) $3! \times 9!$



Total Circular Arrangements

$$= (9-1)! \times 2!$$

$$= (b)$$

(X) $P(\text{rob}) = ? = \frac{\text{fav}}{\text{Total}} = \frac{8! \times 2!}{(10-1)!} = \frac{2}{9}$ Ans

A and B stand in a ring with 10 other persons. If the arrangement of the 12 person is at random, find the chance that there are exactly 3 persons between A and B.

(a) 2/11

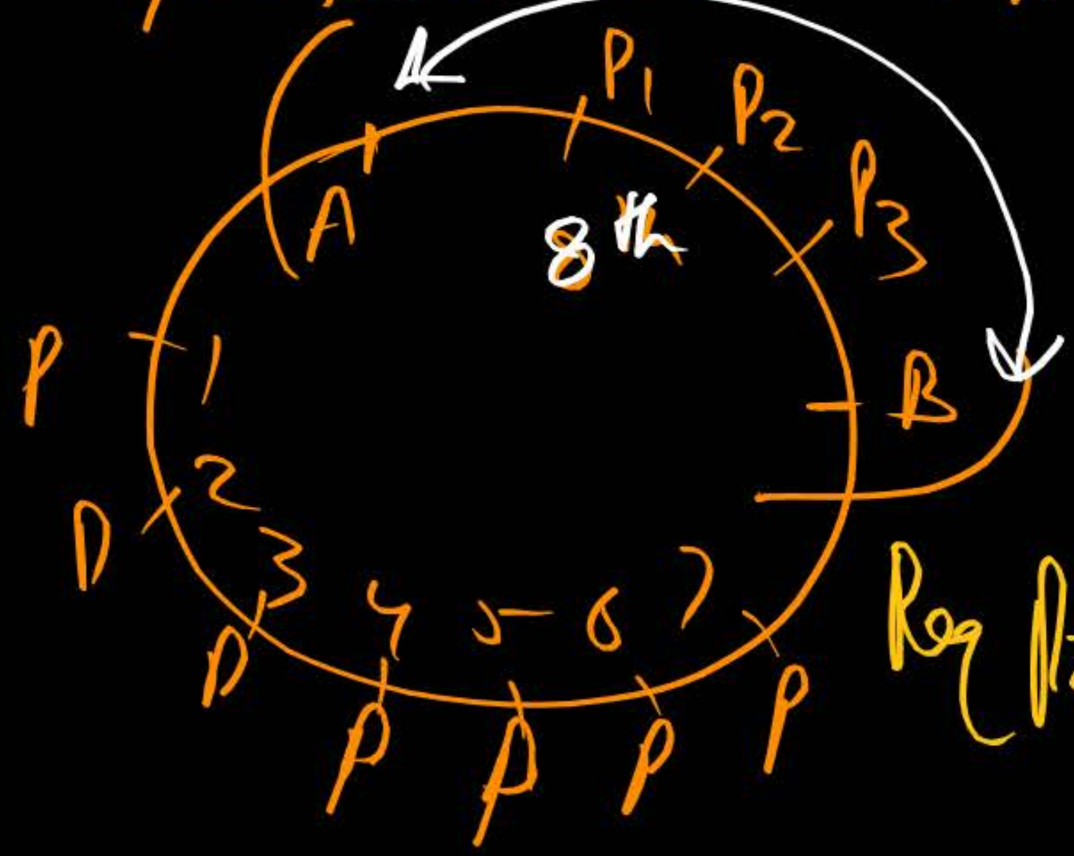
(b) 1/11

(c) 2/11!

(d) 2/9!

Total Persons = 12

So Total circular Arrangements = $(12-1)! = 11!$



fav arrangements =

$$= \left({}^{10}C_3 \times 3!\right) \times (8-1)! \times 2!$$

$$\text{Req Prob} = \frac{\text{fav}}{\text{Total}} = \frac{{}^{10}C_3 \times 3! \times 7! \times 2!}{11!} = \frac{2}{11}$$

(a) ✓

Permutation of Alike items : →

(i) No. of linear arrangements of n different things $= n!$ (RNA)

(ii) No of Circular " " n different things $= (n-1)!$ (RNA)

(iii) No of Linear arrangement of n things in which p are alike, q are alike, r are alike & rest are different $= \dfrac{n!}{p!\,q!\,r!}$ RNA

where $p+q+r+$ Rest items $=$ Total items
(?) (n)

Q. How many different 11 letter words can be formed using the letters of the word 'INEFFECTIVE'

Sol: $\underbrace{I, I}_{p=2}, \underbrace{E, E, E}_{q=3}, \underbrace{F, F}_{r=2}, H, C, T, V$ & Total $(n) = 11$

So Total 11 letter words $= \dfrac{11!}{2!.3!.2!}$

Plastic Toys (RNA)

Q. How many different 9 letter words can be formed using the letters of the word 'ALLAHABAD'

Sol: $\underbrace{A, A, A, A}_{p=4}, \underbrace{L, L}_{q=2}, H, B, D$

Req Ans $= \dfrac{9!}{4! \times 2!}$

A library has 'a' copies of one book, 'b' copies of each of two books, 'c' copies of each of three books and single copies of 'd' books, then the total number of ways in which these books can be arranged are? single

(a) $\frac{(a+b+c+d)!}{a!b!c!}$ (b) $\frac{(a+2b+3c+d)!}{a!(b!)^2(c!)^3}$ (c) $\frac{(a+2b+3c+d)!}{a!b!c!}$ (d) None

Sol

$\underbrace{EEE---E}_{a \text{ copies}},\ \underbrace{HHH---H}_{b \text{ copies}},\ \underbrace{M,M,M--M}_{b \text{ copies}},\ \underbrace{S,S--S}_{c \text{ copies}},\ \underbrace{G,G,G--G}_{c \text{ copies}},\ \underbrace{PPP--P}_{c \text{ copies}},\ \underbrace{X,Y,....Z}_{d \text{ Books.}}$

Total Books $=(a+2b+3c+d)$ books $= n$

Req Arrangements $= \dfrac{(a+2b+3c+d)!}{(a!)(b!)(b!)(C!)(C!)(C!)}$

(b) ✓

If the letters of the word "MOTHER" are written in all possible orders and then written out as in a dictionary then rank of the word MOTHER is

(a) 240 (b) 261 (c) 308 (d) 309

Sol.

Total arrangements of 6 letters $= 6! = 720$ words.

E, H, M, O, R, T (RNA)

words starting with E _ _ _ _ _ $= 5! = 120$ words

" " " H _ _ _ _ _ $= 5! = 120$ "

" " " ME _ _ _ _ $= 4! = 24$ "

" " " MH _ _ _ _ $= 4! = 24$ "

$\}$ 288

E, H, M, O, R, T

words starting with MOE _ _ _ = 3! = 6

" " " MOH _ _ _ = 3! = 6

" " " MOR _ _ _ = 3! = 6

" " " MOTE _ _ = 2! = 2

" " " MOTHER = 1 word

$$\left.\begin{matrix}6\\6\\6\\2\\1\end{matrix}\right\} 21$$

ie Rank = 288 + 21 = 309th

(d)

E, H, M, O, R, T

E — 5! = 120

H — 5! = 120

M
- ME = 4!
- MH = 4!
- MO
  - MOE = 3!
  - MOH = 3!
  - MOR = 3!
  - MOT
    - MOTE — 2!
    - MOTHER = 1
    - No Need

309

PUNEET

= ??

NIRMAN
2025

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Thank

YOU